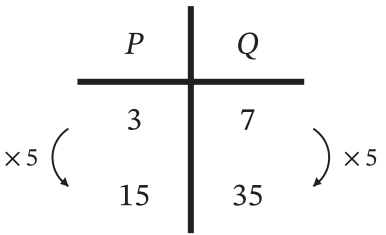
9 1 litre of a liquid **P** has a mass of p grams.
1 litre of a liquid **Q** has a mass of q grams.

A liquid **R** is made by mixing a volume of liquid **P** with a volume of liquid **Q** in the ratio 3 : 7

Find an expression, in terms of p and q, for the mass of 50 litres of liquid **R**.

Total parts = 3 + 7 = 10; one part = 50 ÷ 10 = 5 litres.

1 mark
Split50 in ratio3:7



P:15 litres; Q:35 litres.

1 mark
15 litres;35 litres

Mass = 15p + 35q

= 5(3p + 7q) grams.

1 mark
5(3p+7q) grams

..... grams

(Total for Question 9 is 3 marks)